

MSE 413 Lab 1 Report

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Background

There are 4 regression models used to predict outcomes depending on the number of independent variables.

- Linear regression finds the linear relationship between one outcome and one input based on a pre-trained set of inputs and outputs.

$$y \approx a + bx_1$$

- Multiple linear regression finds the linear relationship between one outcome based on various inputs. Each input has a distinct impact on predicting the outcome.

$$y \approx a + a_1x_1 + a_2x_2 + \dots + a_nx_n$$

- Polynomial regression is a generalized linear regression when an outcome has a polynomial dependence on one input.

$$y \approx a + a_1x_1 + a_2x_2^2 + \dots + a_nx_n^n$$

- Non-linear regression is applied when linear regression models are not fit for predicting the outcomes, as the relationship between input and outcome is not linear or polynomial relationships.

Objectives

- Explore the whole workflow of a machine-learning problem
- Familiarize yourself with Python's machine learning-related libraries such as Pandas, NumPy and Scikit-Learn
- Familiarize with data visualization with Python libraries, such as Seaborn and Matplotlib

Main steps of a machine learning project

1. Frame the problem: Identify inputs and outputs, type of problem (supervised/unsupervised), and type of task (regression, classification, etc)
2. Get the data
3. Discover and visualize the data to get insight
4. Prepare the data for a machine-learning algorithm
5. Select a model and train it on a training dataset
6. Performance measurement and model tuning to eliminate any overfitting
7. Model generalization

Data Discovery

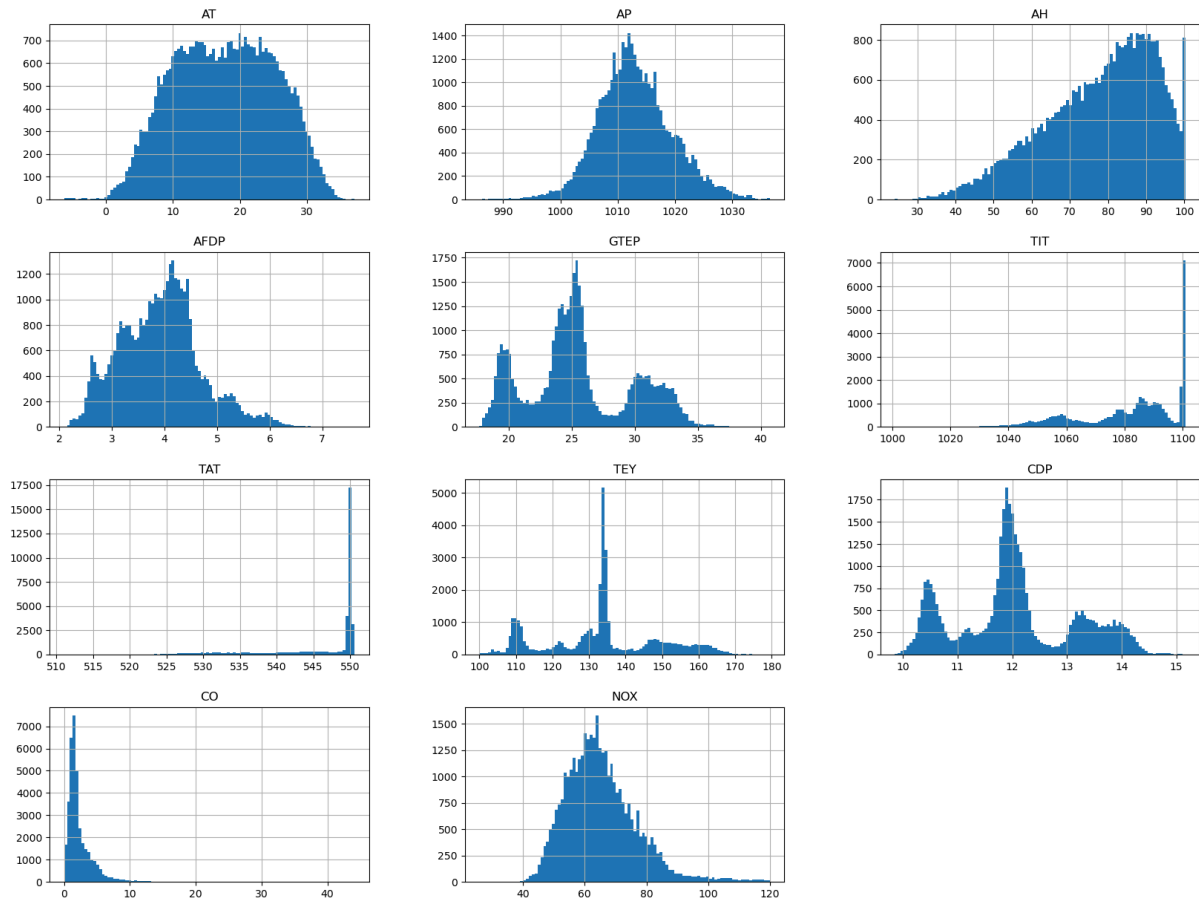


Figure 1: Histograms of all attributes in the dataset

- The data is organized in a single data frame for statistical analysis
- The data is consistent and properly formatted. There are no missing values or null values in the dataset
- The skewness is decided by comparison of mean and mode. Positive skewness (mean > mode): AT, AH, TIT, TAT. Negative skewness (mean < mode) AP, AFDP, GTEP, TEY, CDP, CO, NOX. Alternatively, use the skew method provided in pandas libraries
- Attributes with normal distribution can be determined by the closeness of mean and mode, including AP and NOX. Semi-normal distribution has peak skew on one side of the graph and a trend of increasing or decreasing frequency, such as AH and ATDP
- Graphs that have an abrupt cut-off at the boundary, a sharp peak at the cap, and asymmetry in the distribution or discontinuity in the tail such as TIT, TAT, TEY and CO (sharp peak), and AH (discontinuity in the tail)
- Graphs with multiple peaks in distribution are multimodal distributions. Multimodal distribution: AT, GTEP, CDP.

For linear-type models in machine learning, the multimodal distribution creates bias, poor clustering performance, overfitting problems in supervised learning and feature dependence issues. To tackle this situation, many nonlinear models do not rely on a single distribution such as decision trees, random forests, and neural networks. Another approach to handle these data is to use mixture models to handle multiple distribution peaks. Feature engineering and transformation are alternative solutions to decluster the data, handpick and finetuning models, in trade for more manual labor.

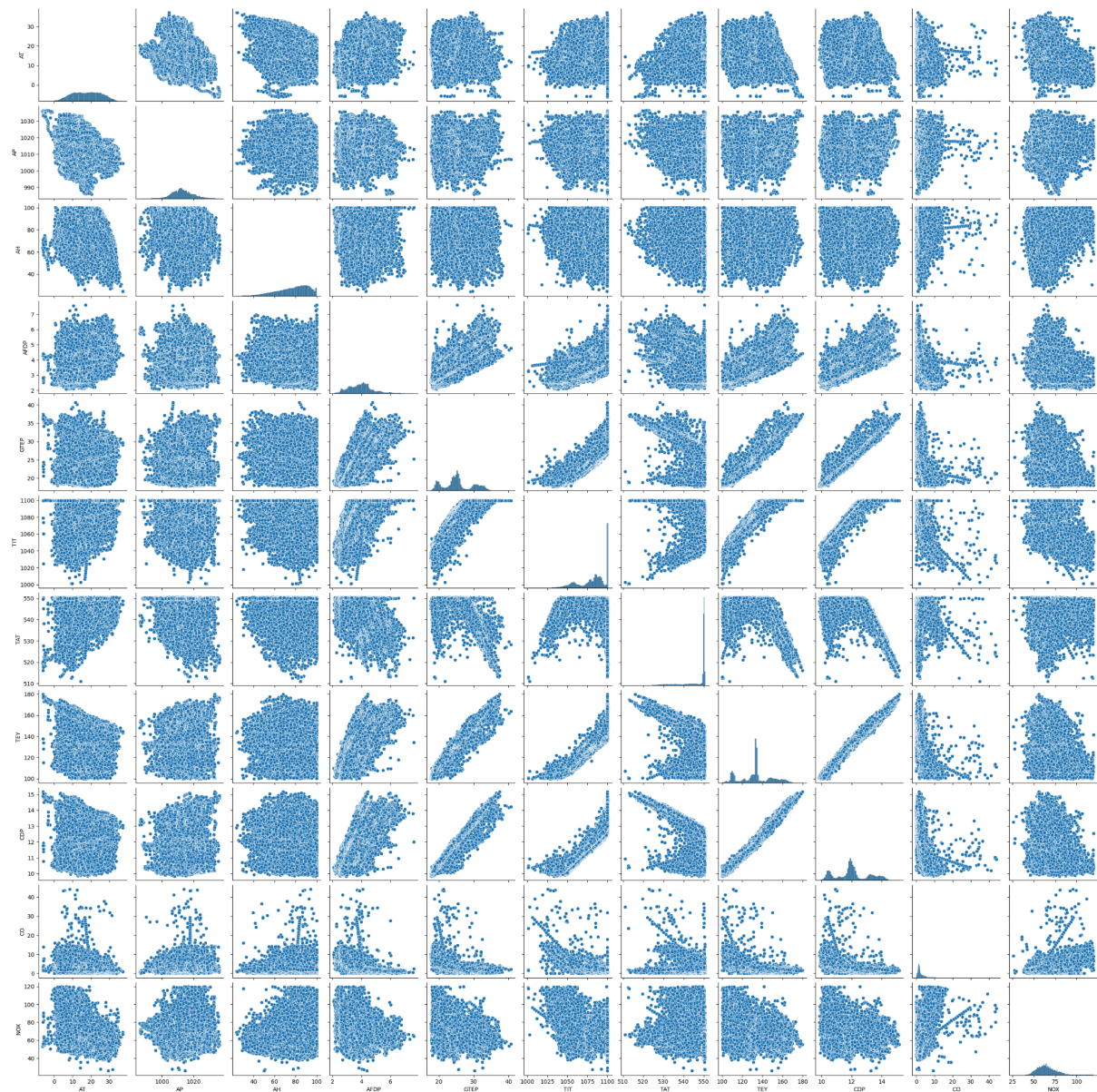


Figure 2: Visualization of all pairs of attributes in the dataset

For TEY

TEY: 1.000000, CDP : 0.988778, GTEP: 0.964127, TIT : 0.910297, AFDP: 0.665483, AP : 0.118224, AT :-0.091152, NOX :-0.116127, AH :-0.137360, CO :-0.569813, TAT :-0.682396

- Positive correlation: CDP, GTEP, TIT, AFDP, AP
- Negative correlation: AT, NOX, AH, CO, TAT

Repeat the steps above for CO and NOX emissions.

For CO

CO: 1.000000, NOX : 0.340606, AH : 0.106586, AP : 0.067050, TAT : 0.058353, AT : -0.174326, AFDP: -0.448425, GTEP: -0.518909, CDP : -0.551027, TEY : -0.569813, TIT : -0.706275

- Positive correlation: NOX, AH, AP, TAT
- Negative correlation: AFDP, GTEP, CDP, TEY, TIT

For NOX

NOX: 1.000000, CO : 0.340606, AP : 0.191938, AH : 0.164617, TAT :-0.092791, TEY :-0.116127, CDP :-0.171256, AFDP:-0.188247, GTEP:-0.201630, TIT :-0.213865, AT :-0.558174

- Positive correlation: CO, AP, AH
- Negative correlation: TAT, TEY, CDP, AFDP, GTEP, TIT, AT

Which feature(s) are highly correlated with all the above target values?

- TEY has positive correlations with CDP, GTEP, and TIT; and a negative correlation to TAT
- CO has a high negative correlation to TIT
- Further view, there is high correlation pair with absolute value greater than 0.7, such as (GTEP, TIT), (TEY, GTEP), (TEY, TIT), (CDP, AFDP), (CDP, GTEP), (CDP, TIT), (CDP, TAT)
- Narrowing down the correlation with 0.7 and 0.9 to avoid the chance of redundant parameters instead of parameters that directly affect the target, the top features are TIT, GTEP, TAT, CDP, AFDP, and CO. Top 4 features are TEY, TIT, GTEP, and CDP.

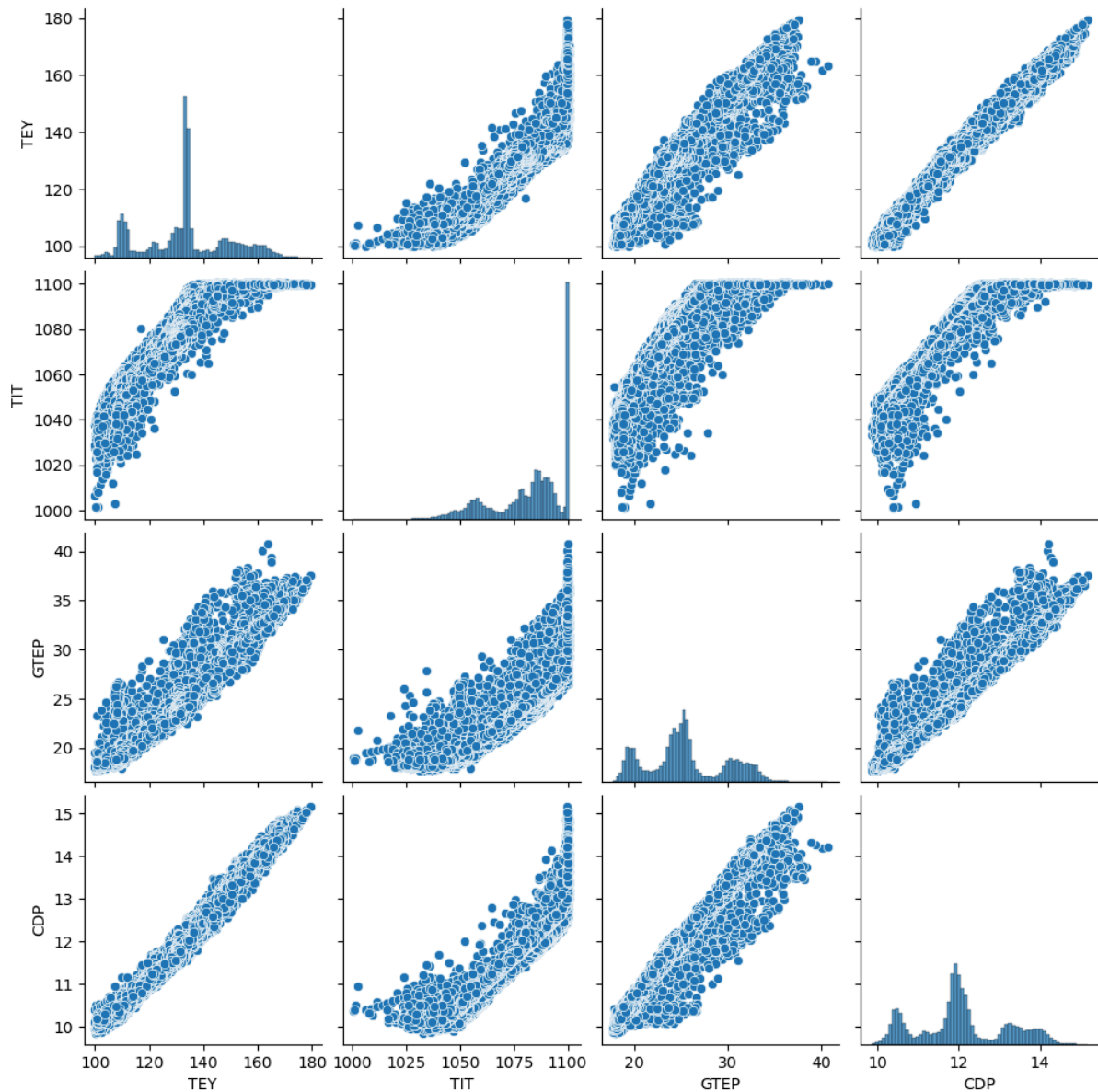


Figure 3: Visualization of hand-picked attributes that have reasonable correlations in pairs

Based on Figure 3, the parameters are pretty linear to each other and simple enough for training and making predictions based on machine-learning techniques. There is no need for the complex relationship of inter-relationship between features.

Simple Linear Regression

For this part of the report, after the data is shuffled and split 80% for training data and 20% data for testing and evaluating the accuracy. The pipeline for building a simple linear regression model is the same. Still, the amount of information fed into the machine for training has a large impact on the accuracy of the data in testing conditions. Attributes can have linear correlations to each other, which makes linear regression a simple task.

Target	Feature	R^2 Train	R^2 Test	RMSE Train	RMSE Test
TEY	CDP	0.9778	0.9771	2.3266	2.3599
CO	TIT	0.5024	0.4857	1.5837	1.6723
NOX	AT	0.3127	0.3068	9.7161	9.5842

Table 1: Model evaluation results for choosing the most correlated attributes to the target.

Root mean squared error (RMSE) indicates the performance of a model. The higher the value, the worse the performance of the model is. In this case, the TEY attribute is highly correlated to CDP (Figure 1), the squared error and RMSE values are relatively low and consistent in both training and testing conditions. The NOX and AT have a very low correlation to each other, which explains why the errors are high in both training and testing data. Co and TIT have a high linear relationship at the beginning but diverge as both attributes inversely increase. The squared errors and RMSE values are surprisingly low for training and testing conditions.

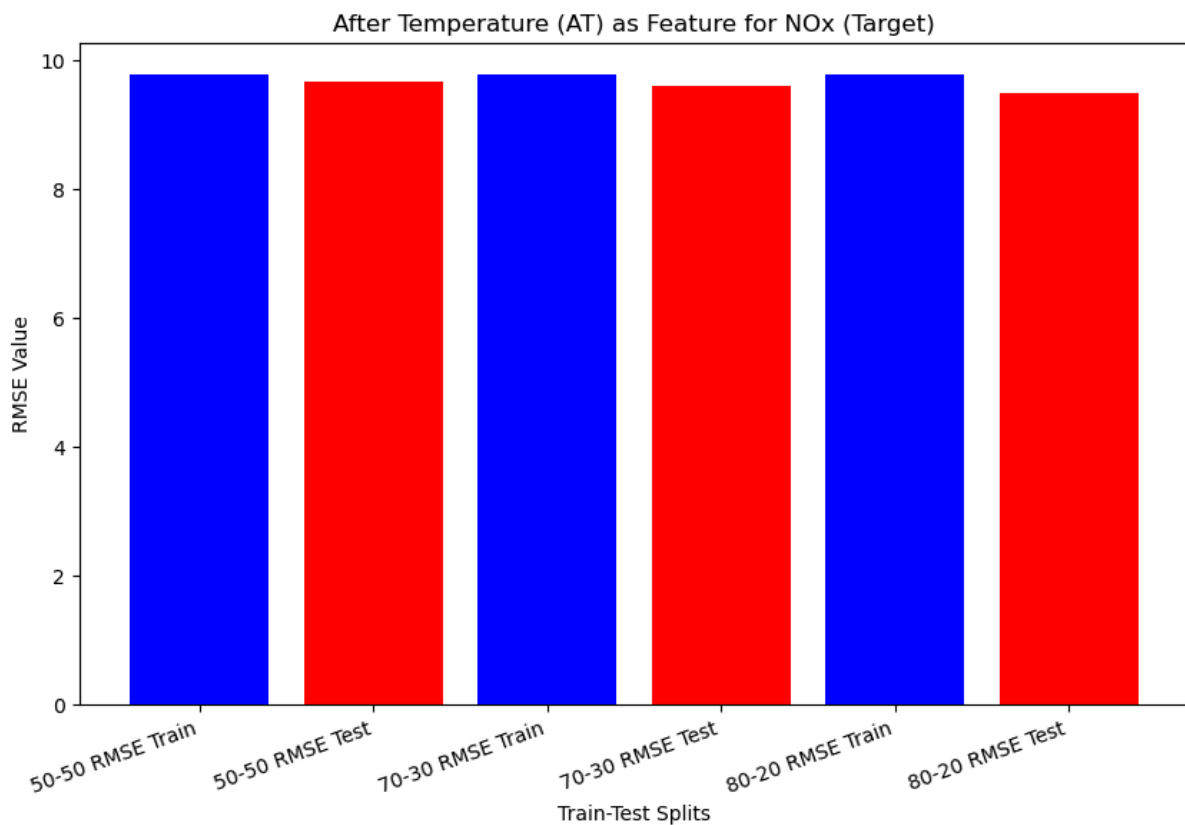


Figure 4: RMSE in training and testing conditions with different data splitting portions of shuffled data

The splitting portion does have an impact on the accuracy of the prediction in both training and testing conditions. The reason is AT and NOX are poorly correlated (figure 1), which significantly contributes to the squared error and RMSE across all splitting portions in both training and testing conditions. The second observation in the above figure is that there is a

downward trend in accuracy when the training portion is increased. This trend has to be put to a test where 2 attributes have a higher correlation.

Multiple Linear Regression

Target	Feature 1	Feature 2	Feature 3	Feature 4
TEY	CDP	GTEP	TIT	TAT
NOX	AT	TIT	GTEP	AP

Table 2: Features that have the highest correlations to the target TEY and NOX based on Pearson's correlation formula

Target	Features	R^2 Train	R^2 Test	RMSE Train	RMSE Test
TEY	CDP, GTEP, TIT, TAT	0.9788	0.9782	2.2756	2.3074
NOX	AT, TIT, GTEP, AP	0.3527	0.3449	9.4291	9.3173

Table 3: Results for multiple linear regression models with the 4 highest correlated features for each target (TEY, NOX)

The highest correlated attributes are picked for this multiple linear regression model. In general, the higher correlated set of attributes to the target archives a better linear regression model and produces less error. Meanwhile, the lower correlated set makes more errors when predicting the target. This is because NOX has almost no relation to any attributes collected in the dataset (Figure 1). The data is highly independent of other attributes.

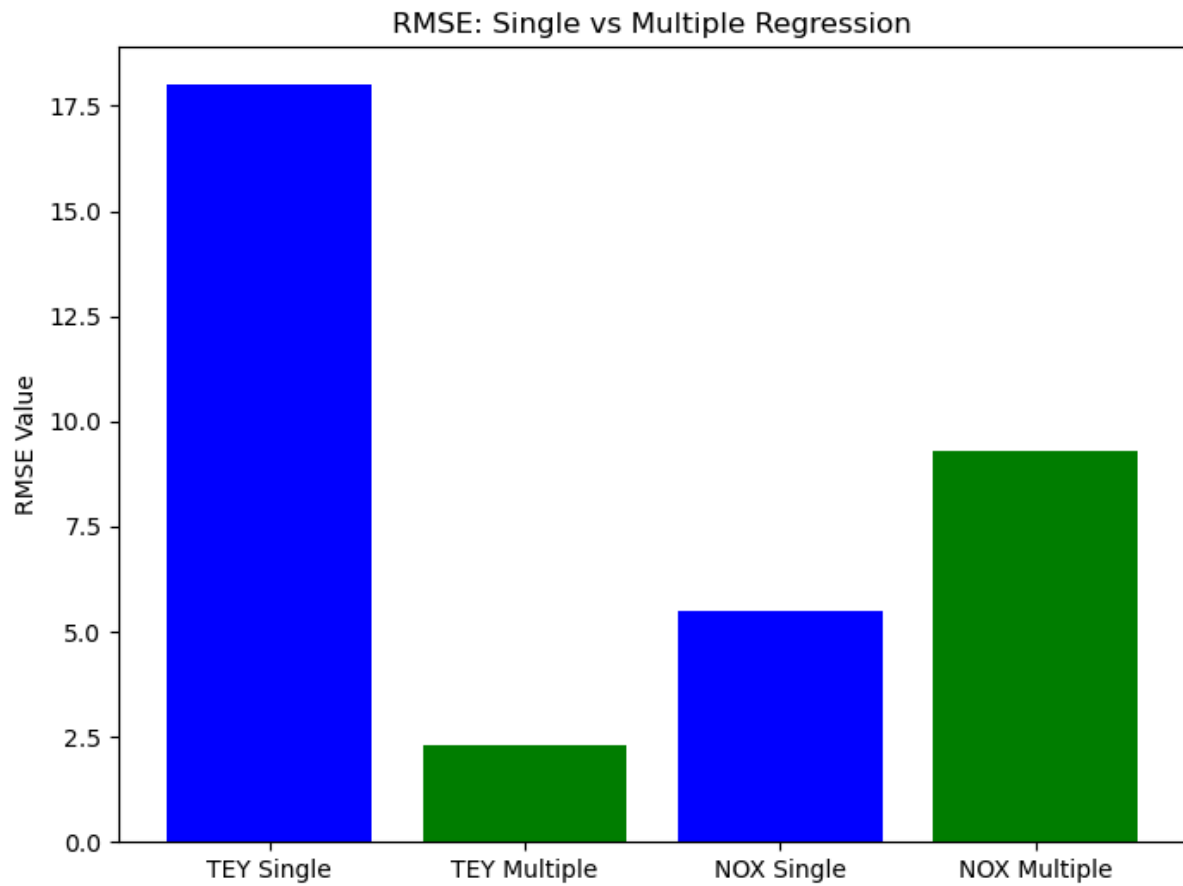


Figure 5: Comparison between single and multiple linear regression for each target

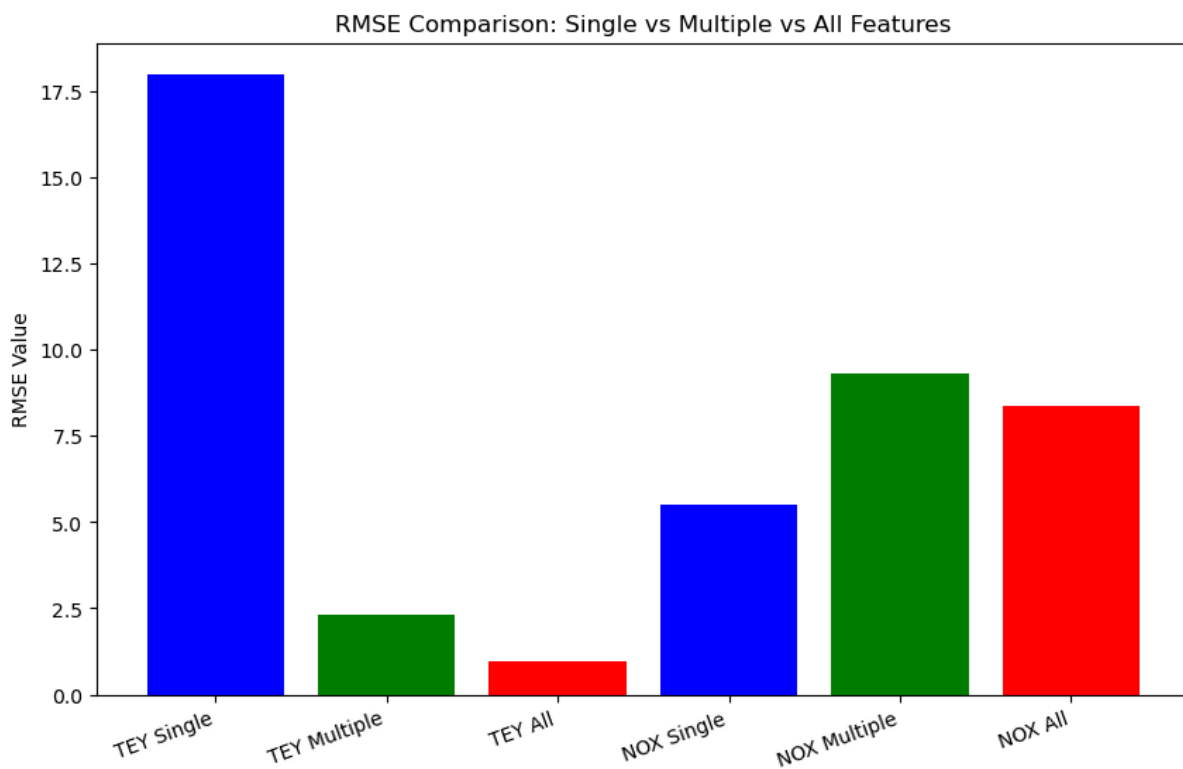


Figure 6: Comparison between single, multiple and all features in linear regression

Does employing more features to train the ML model improve performance?

- A strong correlation with the target data does not necessarily mean better performance, since each feature can constrain others in getting to the actual result. There is a general trend where highly correlated data can predict more accurate data. It depends on the dependent nature of the target attribute, concerning other attributes recorded in the dataset.
- In the TEY case, the parameter is highly dependent on several factors. This makes predicting solely based on the known TEY data impossible. Incorporating highly correlated parameters in predicting significantly improves the accuracy because it matches the nature of this attribute. Incorporating all features makes an even better prediction because the TEY is the output of all other attributes.
- In the NOX case, the situation is the opposite of TEY. NOX is an independent parameter. It is very likely to be the input of the system dataset, as the highest accuracy for prediction is only based on the NOX data itself. The value of NOX becomes wildly incorrect when factoring in more attributes, assuming they all directly affect the predicted values.

Polynomial Regression

Model	R^2 Train	R^2 Test	RMSE Train	RMSE Test
Degree 2 polynomial	0.7335	0.7285	6.0494	5.9983
Degree 3 polynomial	0.7953	0.7878	5.3021	5.3032
Degree 5 polynomial	0.8642	0.6923	4.3180	6.3851
Tree decision regression	1.0000	0.7389	0.0000	5.8825

Table 4: Results of polynomial regression with different degrees of freedom and tree decision regression model

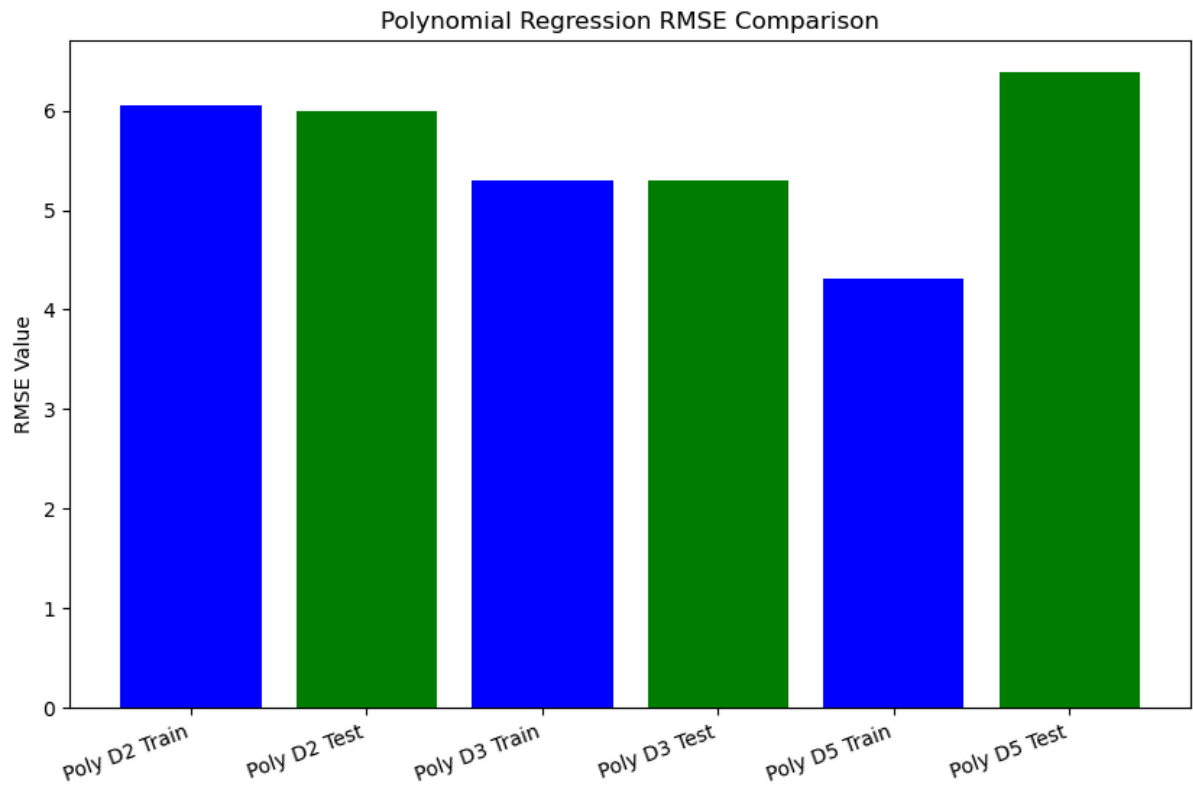


Figure 7: Comparison between different degrees of polynomial regression model

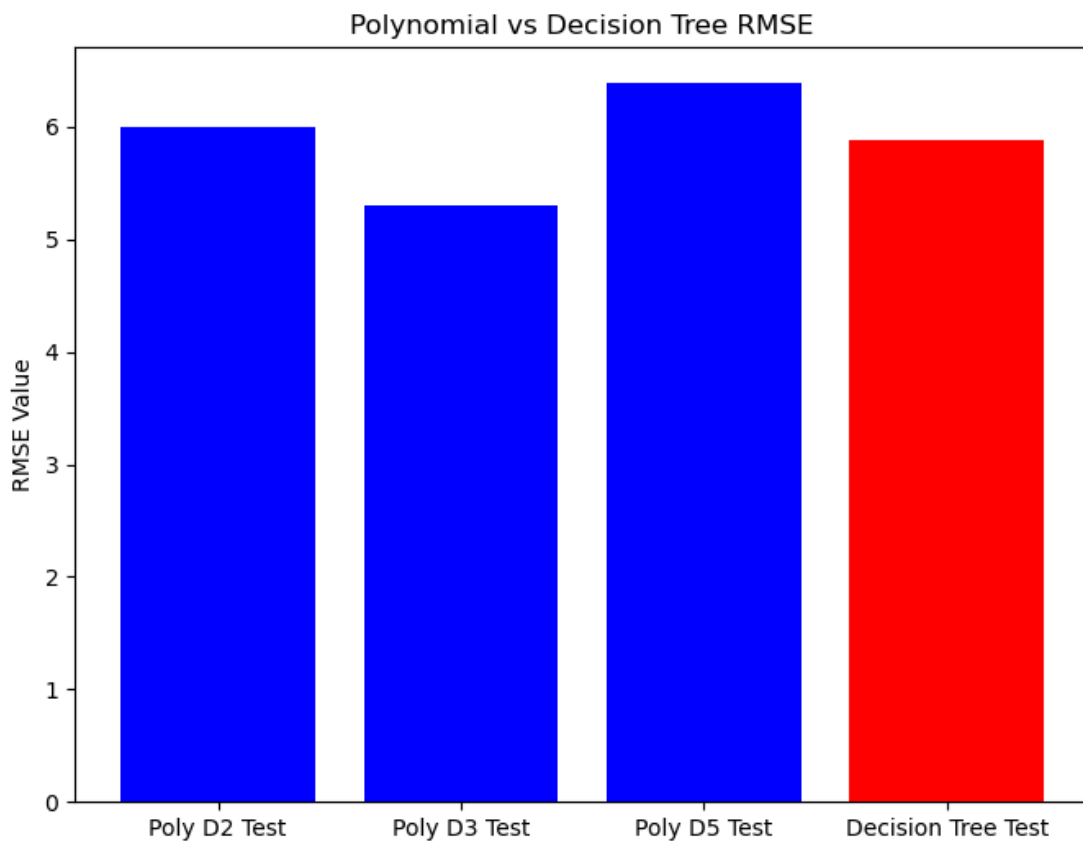


Figure 8: Comparison of RMSE value between different degrees of polynomial regression and decision tree regression

Tree decision and 5th-degree polynomial regression are overfitting, based on the huge discrepancy between squared error (bias) and RMSE value (variance) in both training and testing conditions. The degree 3 polynomial model hits the sweet spot for value prediction as the bias and variance are consistent and not sensitive to changes in other parameters. The degree 2 polynomial is also a balanced performance with less bias variance, in trade of more variance. Both degrees 2 and 3 are good models to consider

In the tutorial, we introduced you to decision tree regressor, as a more advanced regression model. Use a decision tree to retrain your model on The Turbine NOX emission. Compare your results with part (a). Compare the performance of the model on train and test data. Is there any overfitting? Do a bit of search and briefly describe which hyperparameters of this algorithm should be manipulated to adjust overfitting.

- The third degree has the lowest RMSE, which means 3 is the sweet spot in polynomial regression for this dataset. The model is a consistent performance in both training and testing conditions at the third degree. The trend is that the higher the degree, the lower the RMSE value in the training, but also more prone to error and overfitting in testing conditions.
- There is a small discrepancy in train degree 2 and 3 polynomial regression in RMSE values. Degree 3 polynomial is the sweet spot for this dataset due to its lowest error in both training and testing performance. The huge discrepancy in RMSE value in training and testing conditions shows the overfitting in the model
- In the code, there is a Ridge variable to adjust the stability of the polynomial. The hyper-parameter in this case is the stability factor of the polynomial regression. There is no general trend of where the polynomial has the lowest RMSE values in both training and testing conditions with low discrepancy between the 2 of them. For the decision tree, the hyperparameter unique to this model is the depth of the decision. Other hyperparameters applied for all training models are splitting between training and testing and choosing the best portion for each section.

Conclusion

In this lab report, we learn to build a pipeline from collecting data, to analyzing and visualizing data for insights about relationships between attributes, feeding data into a regression model and evaluating the bias and variance in the model's prediction capabilities. Cleaning data and investigating the relationship between attributes are important as they are important later on when performing feature engineering and finetuning models. The next step is choosing models to make specific target attributes. There is no common rule to decide what set of functions can be balanced between bias and variance while giving an acceptable prediction. The same thing applies to choosing hyperparameters for finetuning non-linear models.